Bahria University, Islamabad Campus

Course: CS-: Discrete Structures
Course Instructor: Engr. Saad Mazhar Khan
Max Marks: 20 | **CLO:** CLO-1
Date: 13th July 2026
Due Date: 19th July 2026

Instructions:

1. **Format:** This assignment must be submitted in handwritten form on A4-sized plain or ruled paper. Digital printouts (except for the cover page) will not be accepted.
2. **Clarity:** Use a **Blue or Black ballpoint pen** for text. Use a **Ruler** to draw the Truth Table to ensure neatness and readability.
3. **Group Identification:** Mention the Names and Roll Numbers of both group members clearly on the top right corner of the first page.
4. **Symbolic Accuracy:** Ensure that logical symbols ($\neg, \wedge, \vee, \rightarrow, \leftrightarrow$) are written clearly to avoid ambiguity during marking.
5. **Page Numbering:** Number each page (e.g., 1/4, 2/4) to ensure the logical flow of the tasks is maintained.
6. **Keyword Requirement:** In your justifications, you must explicitly use the keyword "SELECT" to demonstrate your decision-making process in mapping the logic.

CASE STUDY: THE "DEEP-PULSE" UNDERSEA CRISIS SYSTEM

Background:

The Deep-Pulse Station is a high-pressure underwater research facility. The station's Life Support & Evacuation System (LSES) is governed by an automated logic controller. Your task is to verify the logical integrity of the evacuation trigger protocol (E) based on four variables:

- P: Internal Hydrostatic Pressure is stable.
- O: Oxygen levels are within the safe threshold.
- R: Reactor core is in a steady state.
- B: Backup life support is active.

The Operational Logic Protocol:

"The Evacuation Trigger (E) is activated if and only if the following interlocking conditions are satisfied: It is necessary that the Internal Hydrostatic Pressure is not stable ($\neg P$) or the Oxygen levels are not safe ($\neg O$) for the trigger to engage. However, the system must adhere to the 'Redundancy Override': the trigger will neither engage if the Reactor core is steady (R) nor will it engage if the Backup life support is functional (B), unless the Internal Hydrostatic Pressure is failing ($\neg P$).

Furthermore, the 'Safety Sufficiency' clause states: If the Reactor core is not steady ($\neg R$), then it is sufficient for the Evacuation Trigger (E) to engage that the Oxygen levels are failing ($\neg O$). Finally, the Engineering Manual dictates that the Backup life support being active (B) is an 'Only If' condition for the Reactor core being steady (R)."

ASSIGNMENT TASKS

Task 1: Formal Symbolic Mapping (5 Marks)

Select and identify the atomic propositions from the text. Translate the entire "Deep-Pulse" protocol into a single, unified complex symbolic expression for E.

- ***Requirement:*** Show the step-by-step translation of phrases like "Necessary," "Neither/Nor" "Sufficient," and "Only If."

Task 2: Logical Variation and Transformation (5 Marks)

Select the "Safety Sufficiency" clause: *"If the Reactor core is not steady ($\neg R$), then it is sufficient for the trigger to engage that the Oxygen levels are failing ($\neg O$)."*

1. Translate this specific clause into symbolic form.
2. Write the **Inverse** of this clause in English.
3. Write the **Converse** of this clause in English.
4. Write the **Contrapositive** of this clause in English.
5. **Select** and justify which of these three variations is logically equivalent to the original clause.

Task 3: Truth Table Construction (8 Marks)

Construct a comprehensive handwritten truth table for the protocol derived in Task 1.

- **Table Size:** $2^4 = 16$ rows (representing all combinations of P, O, R, B).
- **Requirement:** Include intermediate columns for all negations and bracketed sub-expressions.
- **Selection:** Circle or highlight the rows where the Evacuation Trigger (E) evaluates to **True**.

Task 4: Logical Critical Thinking (2 Marks)

Select a scenario from your truth table where the Reactor is failing ($\neg R$) and Oxygen is failing ($\neg O$), yet the Evacuation Trigger (E) remains **False**.

- Briefly explain why the logic prevented the evacuation in this case.
- State whether this logic represents a safe "Redundancy" or a dangerous "Deadlock."